

Summary

Performance-focused Software Engineer with 6.5 years of experience, including 3 years in MLOps, AI reliability, and automated inference systems. Currently pursuing an MS in Computer Science (AI specialization). Proven expertise in end-to-end ML pipelines, RAG optimization, and real-time edge inference, with a record of reducing deployment latency by 40% and improving system stability at enterprise scale. Currently pursuing MS in CS (AI Specialization).

Education

Cleveland State University Master of Science in Computer Science Coursework – Data Structures and Algorithms, Data Mining and Machine Learning, Deep Learning, Computer Vision, Computer Architecture.	Jan. 2025 – Dec. 2026 Cleveland, Ohio
Anna University Bachelor of Engineering in Electronics and Instrumentation Coursework –Java, Python, Front end, Data Structures and Algorithm.	Aug. 2014 – June 2018 Chennai, India

Technical Skills

Languages: Python (Advanced: Pandas, NumPy, Pydantic), SQL, Java
Machine Learning & AI: PyTorch, YOLOv8, LangGraph, Transformers (Hugging Face), RAGAS.
MLOps & Cloud: Docker, Kubernetes (Minikube/K3s), Jenkins, GitHub Actions (CI/CD), MLflow, Prometheus
Data Engineering: Apache Spark, Pinecone, Airflow, Data Validation (Great Expectations)
Methodologies: CI/CD for ML (CT/CD), Agile, Model Governance, Automated Evaluation (RAGAS, TruLens), Microservices Architecture.

Experience

<u>SYNAMEEDIA</u> – AI/ML Engineer	Nov 2022 – Dec 2024
- Architected petabyte-scale video telemetry and logging pipelines utilizing AWS S3 and Snowflake, successfully migrating legacy PostgreSQL stores to a decoupled OLAP architecture; reduced query latency for ML feature extraction by 50%. - Designed and deployed highly scalable RESTful APIs utilizing FastAPI and Kubernetes to serve real-time anomaly detection models for the EverGuard anti-piracy platform, enabling high-throughput identification of unauthorized credential sharing. - Orchestrated end-to-end MLOps workflows and ETL data pipelines utilizing Apache Airflow (replacing legacy Cron infrastructure), ensuring reliable dependency management, automated model retraining, and zero-downtime inference deployments. - Collaborated with cross-functional data science and security teams to operationalize Server-Side forensic watermarking (ContentArmor) models, directly contributing to a 75% faster disruption time for illegal live broadcast streams.	

<u>GARTNER</u> - Software Engineer	Nov 2021 – Sept 2022
- Developed and maintained robust data validation pipelines using PySpark within a Databricks environment, transitioning legacy QA scripts into scalable data engineering workflows to support market intelligence products - Engineered dynamic data quality and market intelligence dashboards utilizing Power BI and Azure Analysis Services, establishing continuous monitoring protocols that reduced manual production support by 20+ hours monthly. - Automated sophisticated ETL workflows utilizing Apache Airflow (replacing legacy Cron schedulers), ensuring high-fidelity data extraction and transformation for downstream analytical reporting and executive dashboards	

<u>ACCENTURE</u> – Associate Software Engineer	Jun 2018 – Oct 2021
- Validated loan process flow logics for a banking client on MainFrame application and created automation test cases using Java selenium. - Created Test plan and Test strategies. Participated in Client calls to understand requirements and plan sprint activities to provide estimation. - Learned to use Application life cycle management tool such as ALM and Test rail. - Added features to existing automation framework such as custom email validation library to increase automation coverage across all cross functional teams. - Completed critical time constrained releases with zero production incident.	

Projects

<u>Underwater Pipeline Defect Detection (Computer Vision)</u> – Cleveland State University
- Developed a real-time structural damage detection system using YOLOv8 and PyTorch, integrating CLAHE-based image enhancement and Alumentations to improve detection accuracy by 15% in turbid underwater conditions. - Tech stack: YOLOv8, PyTorch, OpenCV.
<u>Intelligent Candidate Screening Agent (GenAI)</u>
- Designed and deployed a multi-agent RAG system using LangGraph to automate resume parsing and ranking, leveraging PineCone for semantic search and a Judge Agent for relevance evaluation, reducing manual screening time by 85%. - Tech stack: LangGraph, PineCone, Docker.

Certifications

- AWS Certified Machine Learning - Specialty (MLS-C01)
- Pytorch Fundamentals in Udemy
- DeepLearning.AI: MLOps Specialization (Coursera)